# Revenue Forecasting

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jknguyen-portfolio.vercel.app/projects/revenue-forecasting
github.com/ndkn-code/vng-revenue-forecasting

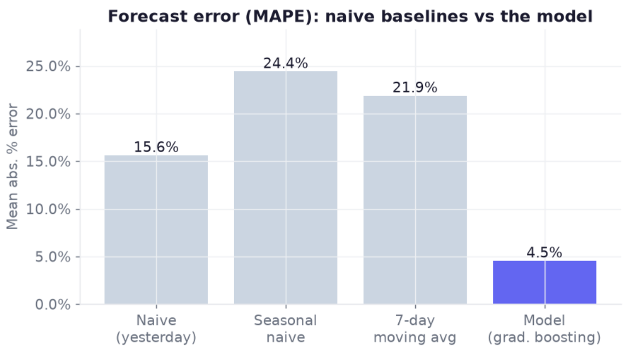
Forecasting / ML · Python · scikit-learn

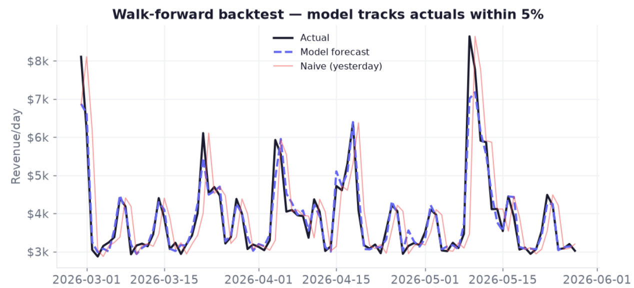
## BUSINESS QUESTION

Forecast a product launch's volatile daily revenue accurately enough to plan budgets against.

## APPROACH

Engineered lag / seasonality / marketing / event features; gradient-boosting model vs naive baselines, scored on a leakage-safe walk-forward backtest. Recreates VNG forecasting work.




## KEY FINDINGS

- Model error 4.5% (MAPE) vs ~22% for a moving-average baseline — a ~80% reduction.
- Naive forecasts fail because they lag launch decay, weekends, and events by a day.
- Recent revenue, marketing spend, and seasonality carry the forecast.

## RECOMMENDATIONS

- Forecast with drivers (marketing, events), not history alone.
- Validate walk-forward, never a random split, to avoid look-ahead leakage.
- A ~4% forecast is tight enough to size UA budgets and cash flow.

Synthetic dataset modeled on a real production schema (confidential). Full case study & code: jknguyen-portfolio.vercel.app/projects/revenue-forecasting · github.com/ndkn-code/vng-revenue-forecasting